



*Automation in modern production systems depend on databases that cannot simply be rolled back like application containers.*

*A single unsafe schema change, large backfill, unindexed foreign key or blocking DDL operation can cause lock contention, application degradation, data inconsistency or prolonged outages. Existing AI coding and database agents primarily generate SQL or execute requested operations; they lack a trustworthy mechanism to **prove that a” production database change is safe before execution.”***

*Our idea **AURA** is built to address this gap through an autonomous, zero-trust database-change rehearsal and verification agent. Its purpose is to transform a natural-language request into an evidence-backed, production-ready change. As an AI agent with LIFO inferential framework, **AURA** discovers the real PostgreSQL schema and workload, generates a multi-phase migration, independently critiques the plan, analyse PostgreSQL lock risks deterministically, and rehearses the migration against an isolated **Neon copy-on-write database branch**. If measured risk exceeds defined thresholds, it automatically replans and rehearses again, with up to three iterations.*

*Build with **TrueForge** as the agent runtime and orchestration layer architect, while TrueFoundry AI Gateway provides unified LLM access and task-based model routing. The system combines LLM reasoning with deterministic PostgreSQL analysis, real rehearsal measurements, schema fingerprinting, immutable evidence and an **HMAC-SHA256 approval gate**. The agent reaches an evidence-backed decision but **stops before production execution**, requiring human authorization through a short-lived 133-second approval token.*

*Real-time applications include zero-downtime migrations, large-table backfills, index and constraint changes, SaaS schema evolution, CI/CD migration gates and high-risk emergency database changes.*

*The core USP is therefore **“Prove before Execute.”** Rather than another SQL-generation copilot, AURA creates measurable evidence that connects AI reasoning to real database behaviour.*

*Its current limitations are rehearsal fidelity, PostgreSQL-specific analysis, workload representativeness and application-level behaviour outside the evidence and tests available to the system.*

*The project GitHub repository access link and demonstration prototype video are enclosed to your perusal.*

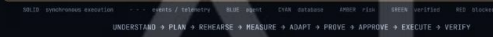
**Team Aura**

Manoj Manu

Karthik M Gowda

Kushal Gowda

Ganesh TN Bhushan



## ZERO-TRUST DATABASE CHANGE CONTROL PLANE / REHEARSE + MEASURE + ADAPT + PROVE



"AURA does not trust the plan. AURA rehearses the plan, measures the result, and proves the change before production."